

Лабораторная работа № 7.

Адресация IPv4 и IPv6. Настройка DHCP

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2024

Цель работы

- ▶ Получение навыков настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6

Настройка DHCP в случае IPv4

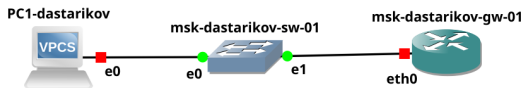


Рис.: Топология моделируемой сети.

Настройка DHCP в случае IPv4

Таблица: Таблица адресации.

Устройство	Интерфейс	IPv4-адрес	Длина префикса	Шлюз по умолчанию	DNS
gw-01	eth0	10.0.0.1	24	-	-
PC1	NIC	Назначен протоколом DHCP	24 (DHCP)	10.0.0.1 (DHCP)	10.0.0.1 (DHCP)

Настройка DHCP в случае IPv4

```
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name dastarikov-gw-01
[edit]
vyos@vyos# set system domain-name dastarikov.net
[edit]
3456@vyos# set system login user dastarikov authentication plaintext-password 12
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
eDone
x[edit]
vyos@vyos# exit
exit
```

Рис.: Настройка маршрутизатора.

Настройка DHCP в случае IPv4

```
dastarikov-gw-01 login: dastarikov
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
dastarikov@dastarikov-gw-01:~$ configure
[edit]
dastarikov@dastarikov-gw-01# delete system login user vyos
[edit]
dastarikov@dastarikov-gw-01# commit
[edit]
dastarikov@dastarikov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
```

Рис.: Настройка маршрутизатора.

Настройка DHCP в случае IPv4

```
dastarikov@dastarikov-gw-01# set interfaces ethernet eth0
↳ address 10.0.0.1/24
dastarikov@dastarikov-gw-01# set service dhcp-server
↳ shared-network-name dastarikov domain-name dastarikov.net
dastarikov@dastarikov-gw-01# set service dhcp-server
↳ shared-network-name dastarikov name-server 10.0.0.1
dastarikov@dastarikov-gw-01# set service dhcp-server
↳ shared-network-name dastarikov subnet 10.0.0.0/24
↳ default-router 10.0.0.1
dastarikov@dastarikov-gw-01# set service dhcp-server
↳ shared-network-name dastarikov subnet 10.0.0.0/24 range
↳ hosts start 10.0.0.2
dastarikov@dastarikov-gw-01# set service dhcp-server
↳ shared-network-name dastarikov subnet 10.0.0.0/24 range
↳ hosts stop 10.0.0.253
dastarikov@dastarikov-gw-01# commit
dastarikov@dastarikov-gw-01# save
dastarikov@dastarikov-gw-01# exit
```

Настройка DHCP в случае IPv4

```
dastarikov@dastarikov-gw-01:~$ show dhcp server statistics
Pool      Size    Leases  Available  Usage
-----
dastarikov 252      0        252      0%
dastarikov@dastarikov-gw-01:~$ show dhcp server leases
IP address  Hardware address  State  Lease start  Lease expiration  Remaining  Pool  Hostname
-----
```

Рис.: Просмотр статистики DHCP-сервера.

Настройка DHCP в случае IPv4

```
Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
Option 12: Host Name = VPCS
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:68:00

Opcode: 2 (REPLY)
Client IP Address: 0.0.0.0
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Offer
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = dastarikov.net
```

Рис.: Настройка DHCP на PC1.

Настройка DHCP в случае IPv4

```
Opcode: 1 (REQUEST)
Client IP Address: 10.0.0.2
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Request
Option 54: DHCP Server = 10.0.0.1
Option 50: Requested IP Address = 10.0.0.2
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:68:00
Option 12: Host Name = VPCS

Opcode: 2 (REPLY)
Client IP Address: 10.0.0.2
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Ack
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = dastarikov.net

IP 10.0.0.2/24 GW 10.0.0.1
```

Рис.: Настройка DHCP на PC1.

Настройка DHCP в случае IPv4

```
VPCS> show ip

NAME           : VPCS[1]
IP/MASK        : 10.0.0.2/24
GATEWAY        : 10.0.0.1
DNS            : 10.0.0.1
DHCP SERVER    : 10.0.0.1
DHCP LEASE     : 86239, 86400/43200/75600
DOMAIN NAME    : dastarikov.net
MAC            : 00:50:79:66:68:00
LPORT          : 20004
RHOST:PORT     : 127.0.0.1:20005
MTU            : 1500

VPCS> ping 10.0.0.1 -c 2

84 bytes from 10.0.0.1 icmp_seq=1 ttl=64 time=1.338 ms
84 bytes from 10.0.0.1 icmp_seq=2 ttl=64 time=1.281 ms
```

Рис.: Проверка IP-адреса на PC1.

Настройка DHCP в случае IPv4

```
dastarikov@dastarikov-gw-01:~$ show dhcp server statistics
Pool      Size      Leases    Available  Usage
-----
dastarikov 252        1         251      0%
dastarikov@dastarikov-gw-01:~$ show dhcp server leases
IP address  Hardware address  State  Lease start      Lease expiration  Remaining  Pool      Hostname
-----
10.0.0.2    00:50:79:66:68:00 active  2025/02/05 18:20:57 2025/02/06 18:20:57 23:56:34  dastarikov  VPCS
```

Рис.: Просмотр статистики DHCP-сервера.

Настройка DHCP в случае IPv4

```
Feb 05 18:20:53 dhcpd[27613]: DHCPDISCOVER from 00:50:79:66:68:00 via eth0
Feb 05 18:20:54 dhcpd[27613]: DHCPOFFER on 10.0.0.2 to 00:50:79:66:68:00 (VPCS) via eth0
Feb 05 18:20:57 dhcpd[27613]: DHCPREQUEST for 10.0.0.2 (10.0.0.1) from 00:50:79:66:68:00 (VPCS) via eth0
Feb 05 18:20:57 dhcpd[27613]: DHCPACK on 10.0.0.2 to 00:50:79:66:68:00 (VPCS) via eth0
```

Рис.: Просмотр логов работы DHCP-сервера.

Настройка DHCP в случае IPv4

628	3158.6389..	0c:e3:05:09:00:00	Broadcast	ARP	60 Who has 10.0.0.2? Tell 10.0.0.1
621	3159.6318..	0.0.0.0	255.255.255.255	DHCP	406 DHCP Discover - Transaction ID 0x1217aa10
622	3159.6410..	10.0.0.1	10.0.0.2	DHCP	342 DHCP Offer - Transaction ID 0x1217aa10
623	3159.6676..	0c:e3:05:09:00:00	Broadcast	ARP	60 Who has 10.0.0.2? Tell 10.0.0.1
624	3168.6917..	0c:e3:05:09:00:00	Broadcast	ARP	60 Who has 10.0.0.2? Tell 10.0.0.1
625	3162.6322..	0.0.0.0	255.255.255.255	DHCP	406 DHCP Request - Transaction ID 0x1217aa10
626	3162.6629..	10.0.0.2	10.0.0.2	DHCP	312 DHCP ACK - Transaction ID 0x1217aa10
627	3163.6324..	00:50:79:66:68:00	Broadcast	ARP	64 Gratuitous ARP for 10.0.0.2 (Request)
628	3164.6336..	00:50:79:66:68:00	Broadcast	ARP	64 Gratuitous ARP for 10.0.0.2 (Request)
629	3165.6344..	00:50:79:66:68:00	Broadcast	ARP	64 Gratuitous ARP for 10.0.0.2 (Request)
630	3335.6975..	00:50:79:66:68:00	Broadcast	ARP	64 Who has 10.0.0.1? Tell 10.0.0.2
631	3335.6990..	0c:e3:05:09:00:00	00:50:79:66:68:00	ARP	60 10.0.0.1 is at 0c:e3:05:09:00:00
632	3335.6996..	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) request id=0x26ad, seq=1/256, ttl=64 (reply in 633)
633	3335.7007..	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) reply id=0x26ad, seq=1/256, ttl=64 (request in 632)
634	3336.7022..	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) request id=0x27ad, seq=2/512, ttl=64 (reply in 636)
635	3336.7031..	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) reply id=0x27ad, seq=2/512, ttl=64 (request in 634)
636	3340.8791..	0c:e3:05:09:00:00	00:50:79:66:68:00	ARP	60 Who has 10.0.0.2? Tell 10.0.0.1
637	3340.8793..	00:50:79:66:68:00	0c:e3:05:09:00:00	ARP	60 10.0.0.2 is at 00:50:79:66:68:00

Рис.: Просмотр перехваченных пакетов.

Настройка DHCP в случае IPv6

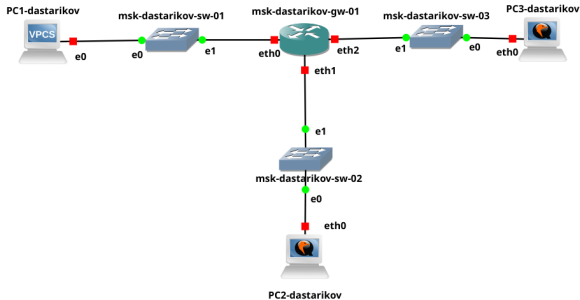


Рис.: Топология моделируемой сети.

Настройка DHCP в случае IPv6

Таблица: Таблица адресации

Устройство	Интерфейс	IPv4-адрес	Длина префикса	Шлюз по умолчанию	DNS
gw-01	eth1	2000::1	64	—	—
gw-01	eth2	2001::1	64	—	—
PC2	NIC	Назначен протоколом SLAAC	64 (SLAAC)	Назначен протоколом SLAAC	
PC2	NIC	Назначен протоколами SLAAC и DHCPv6	64	Назначен маршрутизатором	2000::1 (DHCPv6)
PC3	NIC	Назначен протоколом DHCPv6	64	Назначен маршрутизатором	2001::1 (DHCPv6)

Настройка DHCP в случае IPv6

```
dastarikov@dastarikov-gw-01:~$ configure
[edit]
dastarikov@dastarikov-gw-01# set interfaces ethernet eth1 address 2000::1/64
[edit]
dastarikov@dastarikov-gw-01# set interfaces ethernet eth2 address 2001::1/64
[edit]
dastarikov@dastarikov-gw-01# show interfaces
    ethernet eth0 {
        address 10.0.0.1/24
        hw-id 0c:e3:05:09:00:00
    }
    ethernet eth1 {
+   address 2000::1/64
        hw-id 0c:e3:05:09:00:01
    }
    ethernet eth2 {
+   address 2001::1/64
        hw-id 0c:e3:05:09:00:02
    }
    loopback lo {
    }
[edit]
dastarikov@dastarikov-gw-01# commit
save[edit]
dastarikov@dastarikov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
```

Рис.: Настройка IPv6-адресов на маршрутизаторе.

Настройка DHCP в случае IPv6

```
dastarikov@dastarikov-gw-01# set service router-advert
↳ interface eth1 prefix 2000::/64
dastarikov@dastarikov-gw-01# set service router-advert
↳ interface eth1 other-config-flag
dastarikov@dastarikov-gw-01# set service dhcpv6-server
↳ shared-network-name dastarikov-stateless
dastarikov@dastarikov-gw-01# set service dhcpv6-server
↳ shared-network-name dastarikov-stateless subnet
↳ 2000::0/64
dastarikov@dastarikov-gw-01# set service dhcpv6-server
↳ shared-network-name dastarikov-stateless common-options
↳ name-server 2000::1
dastarikov@dastarikov-gw-01# set service dhcpv6-server
↳ shared-network-name dastarikov-stateless common-options
↳ domain-search dastarikov.net
dastarikov@dastarikov-gw-01# commit
dastarikov@dastarikov-gw-01# save
dastarikov@dastarikov-gw-01# run show configuration
```

Настройка DHCP в случае IPv6

```
root@kali:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 2000::1501:a379:1276:1002 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::519:316b:1f96:f13e prefixlen 64 scopeid 0x20<link>
    ether 0c:7a:f1:34:00:00 txqueuelen 1000 (Ethernet)
    RX packets 21 bytes 2252 (2.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 61 bytes 6829 (6.6 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Рис.: Проверка настроек сети узла PC2.

Настройка DHCP в случае IPv6

```
root@kali:~# route -n -A inet6
Kernel IPv6 routing table

```

Destination	Next Hop	Flag	Met	Ref	Use	If
::1/128	::	U	256	2	0	lo
2000::/64	::	U	100	3	0	eth0
fe80::/64	::	U	100	1	0	eth0
::/0	fe80::ee3:5ff:fe09:1	UG	100	1	0	eth0
::1/128	::	UAn	0	5	0	lo
2000::1501:a379:1276:1002/128	::	UAn	0	3	0	eth0
fe80::519:316b:1f96:f13e/128	::	UAn	0	3	0	eth0
ff00::/8	::	U	256	4	0	eth0
::/0	::	!n	-1	1	0	lo

```
root@kali:~# ping 2000::1 -c 2
PING 2000::1(2000::1) 56 data bytes
64 bytes from 2000::1: icmp_seq=1 ttl=64 time=5.14 ms
64 bytes from 2000::1: icmp_seq=2 ttl=64 time=2.66 ms

--- 2000::1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 2.656/3.895/5.135/1.239 ms
```

Рис.: Проверка подключения к маршрутизатору PC2.

Настройка DHCP в случае IPv6

```
root@kali:~# cat /etc/resolv.conf
search dastarikov.net.
nameserver 2000::1
root@kali:~# dhclient -6 -S -v eth0
Internet Systems Consortium DHCP Client 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on Socket/eth0
Sending on Socket/eth0
Created duid "\000\003\000\001\014z\3614\000\000".
PRC: Requesting information (INIT).
XMT: Forming Info-Request, 0 ms elapsed.
XMT: Info-Request on eth0, interval 1010ms.
RCV: Reply message on eth0 from fe80::ee3:5ff:fe09:1.
PRC: Done.
```

Рис.: Проверка настроек DNS и DHCPv6 узла PC2.

Настройка DHCP в случае IPv6

```
dastarikov@dastarikov-gw-01# run show dhcpv6 server leases
IPv6 address  State  Last communication  Lease expiration  Remaining  Type  Pool  IAID_DUID
-----
[edit]
```

Рис.: Просмотр статистики DHCP-сервера.

Настройка DHCP в случае IPv6

```

- DHCPv6
  Message type: Advertise (2)
  Transaction ID: 0xb3c7c1
  - Identity Association for Non-temporary Address
    Option: Identity Association for Non-temporary Address (3)
    Length: 60
    IAID: f1340000
    T1: 0
    T2: 0
  - Status code
    Option: Status code (13)
    Length: 44
    Status Code: NoAddrAvail (2)
    Status Message: No addresses available for this interface.
  - Client Identifier
    Option: Client Identifier (1)
    Length: 14
    DUID: 000100012f3680950c7af1340000
    DUID Type: link-layer address plus time (1)
    Hardware type: Ethernet (1)
    DUID Time: Feb  5, 2025 23:03:33.000000000 MSK
    Link-layer address: 0c:7a:f1:34:00:00
    Link-layer address (Ethernet): 0c:7a:f1:34:00:00 (0c:7a:f1:34:00:00)
```

Рис.: Перехваченный DHCPv6 пакет.

Настройка DHCP в случае IPv6

```
dastarikov@dastarikov-gw-01# set service router-advert
↳ interface eth2 managed-flag
dastarikov@dastarikov-gw-01# set service dhcpv6-server
↳ shared-network-name dastarikov-stateful
dastarikov@dastarikov-gw-01# set service dhcpv6-server
↳ shared-network-name dastarikov-stateful subnet
↳ 2001::0/64
dastarikov@dastarikov-gw-01# set service dhcpv6-server
↳ shared-network-name dastarikov-stateful subnet
↳ 2001::0/64 name-server 2001::1
dastarikov@dastarikov-gw-01# set service dhcpv6-server
↳ shared-network-name dastarikov-stateful subnet
↳ 2001::0/64 domain-search dastarikov.net
dastarikov@dastarikov-gw-01# set service dhcpv6-server
↳ shared-network-name dastarikov-stateful subnet
↳ 2001::0/64 address-range start 2001::100 stop 2001::199
dastarikov@dastarikov-gw-01# commit
dastarikov@dastarikov-gw-01# save
```


Настройка DHCP в случае IPv6

```
dastarikov@dastarikov-gw-014 run show dhcpv6 server leases
```

IPv6 address	State	Last communication	Lease expiration	Remaining	Type	Pool	IAID_GUID
2001::198	active	2025/02/05 20:20:09	2025/02/05 22:25:09	2:04:38	non-temporary	dastarikov-stateful	00:00:af:04:00:04:2c:a5:da:e5
:9c:cb:ba:34:91:d4:21:f4:51:0f:36:41							

Рис.: Просмотр статистики DHCP-сервера.

Настройка DHCP в случае IPv6

```
root@kali:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::b9d0:541b:10b:27 prefixlen 64 scopeid 0x20<link>
    inet6 2001::198 prefixlen 128 scopeid 0x0<global>
    ether 0c:2c:04:af:00:00 txqueuelen 1000 (Ethernet)
    RX packets 11 bytes 1312 (1.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 296 bytes 45946 (44.8 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Рис.: Проверка настроек сети.

Настройка DHCP в случае IPv6

```
root@kali:~# route -n -A inet6
Kernel IPv6 routing table
Destination                Next Hop                    Flag Met Ref Use If
::1/128                    ::                          U    256 2    0 lo
2001::198/128              ::                          U    100 1    0 eth0
fe80::/64                  ::                          U    100 1    0 eth0
::/0                       fe80::ee3:5ff:fe09:2       UG   100 3    0 eth0
::1/128                    ::                          UAn  0   5    0 lo
2001::198/128              ::                          UAn  0   3    0 eth0
fe80::b9d0:541b:10b:27/128 ::                          UAn  0   3    0 eth0
ff00::/8                   ::                          U    256 4    0 eth0
::/0                       !n    -1 1    0 lo
```

Рис.: Проверка настроек сети.

Настройка DHCP в случае IPv6

```
Listening on Socket/eth0
Sending on Socket/eth0
Created duid "\000\001\000\001\0\2050\014,\004\257\000\000".
PRC: Soliciting for leases (INIT).
KMT: Forming Solicit, 0 ms elapsed.
KMT: X-- IA_NA 04:af:00:00
KMT: | X-- Request renew in +3600
KMT: | X-- Request rebind in +5400
KMT: Solicit on eth0, interval 1050ms.
RCV: Advertise message on eth0 from fe80::ee3:5ff:fe09:2.
RCV: X-- IA_NA 04:af:00:00
RCV: | X-- Starts 1738787024
RCV: | X-- t1 - renew +0
RCV: | X-- t2 - rebind +0
RCV: | X-- [Options]
RCV: | | X-- IAADDR 2001::199
RCV: | | | X-- Preferred lifetime 27000.
RCV: | | | X-- Max lifetime 43200.
RCV: X-- Server ID: 00:01:00:01:2f:36:6f:49:0c:e3:05:09:00:01
RCV: Advertisement recorded.
PRC: Selecting best advertised lease.
PRC: Considering best lease.
PRC: X-- Initial candidate 00:01:00:01:2f:36:6f:49:0c:e3:05:09:00:01 (s: 10105, p: 0).
KMT: Forming Request, 0 ms elapsed.
KMT: X-- IA_NA 04:af:00:00
KMT: | X-- Requested renew +3600
KMT: | X-- Requested rebind +5400
KMT: | | X-- IAADDR 2001::199
KMT: | | | X-- Preferred lifetime +7200
KMT: | | | X-- Max lifetime +7500
KMT: V IA_NA appended.
KMT: Request on eth0, interval 970ms.
RCV: Reply message on eth0 from fe80::ee3:5ff:fe09:2.
RCV: X-- IA_NA 04:af:00:00
RCV: | X-- Starts 1738787025
RCV: | X-- t1 - renew +0
RCV: | X-- t2 - rebind +0
RCV: | X-- [Options]
RCV: | | X-- IAADDR 2001::199
RCV: | | | X-- Preferred lifetime 7200.
RCV: | | | X-- Max lifetime 7500.
RCV: X-- Server ID: 00:01:00:01:2f:36:6f:49:0c:e3:05:09:00:01
PRC: Bound to lease 00:01:00:01:2f:36:6f:49:0c:e3:05:09:00:01.
```

Рис.: Получение адреса по DHCPv6.

Настройка DHCP в случае IPv6

```
root@kali:~# route -n -A inet6
Kernel IPv6 routing table
```

Destination	Next Hop	Flag	Met	Ref	Use	If
::1/128	::	U	256	2	0	lo
2001::198/128	::	U	100	2	0	eth0
2001::199/128	::	U	256	1	0	eth0
fe80::/64	::	U	100	1	0	eth0
::/0	fe80::ee3:5ff:fe09:2	UG	100	3	0	eth0
::1/128	::	UAn	0	5	0	lo
2001::198/128	::	UAn	0	4	0	eth0
2001::199/128	::	UAn	0	2	0	eth0
fe80::b9d0:541b:10b:27/128	::	UAn	0	3	0	eth0
ff00::/8	::	U	256	4	0	eth0
::/0	::	!n	-1	1	0	lo

Рис.: Просмотр настроек сети.

Настройка DHCP в случае IPv6

```
root@kali:~# ping 2001::1 -c 2
PING 2001::1(2001::1) 56 data bytes
64 bytes from 2001::1: icmp_seq=1 ttl=64 time=7.82 ms
64 bytes from 2001::1: icmp_seq=2 ttl=64 time=2.33 ms

--- 2001::1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 2.333/5.077/7.821/2.744 ms
```

Рис.: Отправка эхо-запроса DHCP-серверу.

Настройка DHCP в случае IPv6

```
root@kali:~# cat /etc/resolv.conf  
search dastarikov.net.  
nameserver 2001::1
```

Рис.: Проверка настроек DNS.

Настройка DHCP в случае IPv6

```
dastarikov@dastarikov-gw-01# run show dhcpv6 server leases
```

IPv6 address	State	Last communication	Lease expiration	Remaining	Type	Pool	IAID_OUID
2001::198	active	2025/02/05 20:20:09	2025/02/05 22:25:09	1:57:53	non-temporary	dastarikov-stateful	00:00:af:04:00:04:2c:a5:da:e5
9c:cb:ba:34:91:d4:21:f4:51:0f:36:41							
2001::199	active	2025/02/05 20:22:59	2025/02/05 22:27:59	2:00:43	non-temporary	dastarikov-stateful	00:00:af:04:00:01:00:01:2f:36
85:4f:8c:12c:04:af:00:00							

Рис.: Просмотр статистики DHCP-сервера.

Настройка DHCP в случае IPv6

```

- DHCPv6
  Message type: Reply (7)
  Transaction ID: 0x40b030
  - Identity Association for Non-temporary Address
    Option: Identity Association for Non-temporary Address (3)
    Length: 40
    IAID: 04af0000
    T1: 0
    T2: 0
  - IA Address
    Option: IA Address (5)
    Length: 24
    IPv6 address: 2001::199
    Preferred lifetime: 7200
    Valid lifetime: 7500
  - Client Identifier
    Option: Client Identifier (1)
    Length: 14
    DUID: 000100012f36854f0c2c04af0000
    DUID Type: link-layer address plus time (1)
    Hardware type: Ethernet (1)
    DUID Time: Feb  5, 2025 23:23:43.000000000 MSK
    Link-layer address: 0c:2c:04:af:00:00
    Link-layer address (Ethernet): 0c:2c:04:af:00:00 (0c:2c:04:af:00:00)
```

Рис.: Перехваченный DHCPv6 пакет.

Выводы

- ▶ В результате выполнения лабораторной работы получили навыки настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6